Tianyi Zhang

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RESEARCH INTERESTS

Trustworthy AI systems, end to end. I study proxy fidelity across the ML pipeline: whether training data reflects the real distribution, learned objectives capture intended behavior, and evaluation metrics track genuine quality and safety. My work spans data and benchmark integrity, representation learning, retrieval evaluation, reward/specification gaming, and monitor robustness.

EDUCATION

Georgia Institute of Technology — Atlanta, GA
Master of Science in Computer Science — *Aug 2025 - Dec 2026*

Emory University — Atlanta, GA
Bachelor of Science in Quantitative Science (Honors) — *Aug 2021 - May 2025*

PROFESSIONAL EXPERIENCE

LinkedIn — Mountain View, CA
AI/ML Engineer Intern | Feed AI, LLM Retrieval — *May 2026 - Present*

- Built a backend-agnostic **Adaptive User Representation Layer** that decouples user understanding from candidate generation. Its compact user-state contract captures short- and long-term intents, lifecycle, confidence/freshness, and evidence provenance as shared conditioning for embedding retrieval and generative recommendation; implemented the first producer with profile/history multi-tower Qwen3 encoding, allowing retrieval backends and model capacity to evolve without rebuilding the user-understanding layer.
- Built and production-scale validated a backward-compatible **PySpark retrieval scorecard over 2.7B member-candidate pairs**, then used matched lifecycle evaluation to correct a **30x population imbalance**; exposed **+25.1% Recall@10 and +26.8% NDCG@10** but **21.6% lower ILD** for Semi-cold versus matched Active members, informing model selection and lifecycle strategy that balance relevance with feed breadth.
- Improved distributed training and model-serving reliability through layered configuration, dry-run validation, reproducible manifests, execution tracking, searchable run catalogs, train/serve parity checks, and serving observability while preserving legacy workflow compatibility; defined one-axis-at-a-time validation across representation, model capacity, retrieval backend, and lifecycle policy.

Efficient & Intelligent Computing Lab (Collaboration w/ NVIDIA Nemotron Research) — Atlanta, GA
Student Researcher | LLM Evaluation, Agentic AI — *Jul 2025 - Dec 2025*

- Co-authored **AgentSuite (ICML 2026)**, a component-based pipeline for auditing LLM-agent benchmarks; surfaced flaws affecting ~23% of tasks across 6 benchmarks and shifting ~63% of model rankings.
- Built taxonomy-guided LLM-judge and structural-detector stages, matching expert labels at **0.79-0.87 F1** and surfacing 100+ undocumented issues.

Emory Graph Mining Lab — Atlanta, GA
Undergraduate Researcher | Graph Learning, AI for Science — *Dec 2023 - May 2025*

- Built a graph-neural-network pipeline over ~8K patients and first-authored a Granger-directed brain-connectivity method at **AIME 2025**.
- Engineered a Transformer-based clinical time-series pipeline modeling transitions across latent functional-connectivity states; oral presentation at **IEEE EMBC 2025**.

PUBLICATIONS

H. Suh*, S. Lee*, B. Ji*, R. Khare, B. Khan, H. Kim, **T. Zhang**, et al. "AgentSuite: Toward More Reliable Agent Evaluation with a Component-Based Benchmark Auditing Pipeline." *International Conference on Machine Learning (ICML)* 2026.

T. Zhang, K. Han, J. Nie, C. You, S. van Rooij, J. Stevens, B. Dunlop, C. Gillespie, C. Yang. "Causal Brain Connectivity: Integrating Granger Directed Graphs in fMRI Analysis." *International Conference on Artificial Intelligence in Medicine (AIME)* 2025. (First Author)

J. Nie, K. Han, **T. Zhang**, C. You, S. van Rooij, J. Stevens, B. Dunlop, C. Gillespie, C. Yang. "Brain Network State Transformer: Leveraging State Functional Connectivity." *IEEE Engineering in Medicine and Biology Society (EMBC)* 2025. (Oral)

SELECTED PROJECTS

CoAct: Governance Substrate for Multi-Agent Coding

- Building a dependency-free Go governance substrate that lets coding agents work safely in one repository through capability policies, advisory locks, worktree isolation, merge gates, a shared task board, and an append-only journal for replayable oversight.

Agent-Loop Observability

- Developing observability tools for diagnosing repeated tool calls, context loss, unrecalled memory writes, cost runaway, and safety-monitor failures in long-running LLM-agent trajectories.

OPEN SOURCE

NVIDIA NeMo RL - Contributor to NVIDIA's scalable reinforcement-learning and LLM post-training library, including checkpointing and response-dataset improvements.

TECHNICAL SKILLS

Programming: Python, Go, SQL, Java, R, C/C++
ML/AI: PyTorch, PyTorch Lightning, CUDA, Hugging Face Transformers, vLLM, SGLang, MLflow
Data/Systems: Spark/PySpark, FSDP, Docker, AWS, GitHub Actions, Linux, PostgreSQL, MongoDB, Redis